

1 Notation

We model four events each year: growth, colonization, fizzle, and end-of-wave impact.

Let (k, t) be the time index, where k represents the year index and t represents the within-year event index: $t = 0$ for the time at the beginning of the year, $t = 1$ for the time after growth, $t = 2$ for the time after colonization, $t = 3$ for the time after fizzle, and $t = 4$ for the time at the end of the year (after the wave). We interpret $(k, 0)$ as the same time as $(k - 1, 4)$.

Assume there are n patches and we label them with p_i ($i = 1, \dots, n$). For each patch we use $(N_i^{(k,t)}, I_i^{(k,t)})$ to describe its state. $N_i^{(k,t)}$ represents population size in patch p_i at time (k, t) , and

$$I_i^{(k,t)} \in \{0, 1, 2, \dots\}$$

is the number of infected hosts in patch p_i at time (k, t) . We define the (derived) infection indicator

$$\mathbb{1}_i^{(k,t)} = \mathbb{1}\{I_i^{(k,t)} \geq 1\}.$$

Let $U \sim \text{Uniform}(0, 1)$ be a random variable uniformly distributed over $(0, 1)$, and we define

$$\mathbb{1}_{U \leq P} = \begin{cases} 1, & \text{if } U \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

2 Parameters

Host:

- r : growth rate
- K : carrying capacity of a patch (all patches are of equal area)
- $c \in [0, 1]$: probability that an individual host moves to another patch during colonization

Epidemic:

- $\mathcal{R}_0$: basic reproduction number
- $P_f = 1/\mathcal{R}_0$: fizzle probability for a *single* introduced infection
- $z = z(\mathcal{R}_0)$: final size fraction of a non-fizzling within-patch epidemic wave
- D : disease-induced mortality fraction among infected hosts (assume $D = 1$)

Propagule (infectious fleas):

- ν : scaling constant linking the previous year's epidemic size in a patch to the total number of infectious propagules generated
- ρ : mean number of infectious propagules that a single migrating live host can carry (carrying capacity per host)

Infection:

- $\beta \geq 0$: mean number of effective bites per propagule during the colonization season
- $p \in [0, 1]$: probability that a single effective bite infects the bitten host
- $\alpha = \beta p$: transmission efficiency (successful infections) per propagule

Propagule assumption We assume free fleas do not survive beyond the current season, so all propagules relevant for year k are generated by the year $k - 1$ wave.

3 Growth

We assume that reproduction of rats follows logistic growth:

$$N_i^{(k,1)} = N_i^{(k,0)} + r N_i^{(k,0)} \left(1 - \frac{N_i^{(k,0)}}{K} \right). \quad (1)$$

Assume infection counts do not change during growth:

$$I_i^{(k,1)} = I_i^{(k,0)} = 0. \quad (2)$$

4 Colonization and Fizzle

4.1 Population size

We assume that each individual rat moves with probability c and lands in a random patch. We ignore distances and assume that all patches are equally available to a moving rat:

$$N_i^{(k,2)} = (1 - c) N_i^{(k,1)} + \frac{c \sum_{j=1}^n N_j^{(k,1)}}{n}. \quad (3)$$

The first term is the population staying in the patch, and the second term is the population moving into the patch.

4.2 Propagules generated by the previous year

Let $\mathbb{1}_j^{(k-1,3)} = \mathbb{1}\{I_j^{(k-1,3)} \geq 1\}$ indicate whether patch p_j is infected after the fizzle step in year $k - 1$. The number of dead and infected hosts produced by the wave in patch j in year $k - 1$ is

$$S_j^{(k-1)} = z D N_j^{(k-1,3)} \mathbb{1}_j^{(k-1,3)}. \quad (4)$$

Propagule capacity and distribution The total expected number of infectious propagules generated in patch j is driven entirely by dead hosts, which is $\nu S_j^{(k-1)}$. However, cross-patch transmission is strictly driven by migrating live hosts. The number of live hosts migrating from patch j is $c N_j^{(k,1)}$. These migrating hosts can carry a maximum expected number of $\rho c N_j^{(k,1)}$ propagules.

We define the expected number of exported propagules $\lambda_{E,j}^{(k)}$ and retained propagules $\lambda_{R,j}^{(k)}$ as:

$$\lambda_{E,j}^{(k)} = \phi \left(\nu S_j^{(k-1)}, \rho c N_j^{(k,1)} \right) \quad (5)$$

$$\lambda_{R,j}^{(k)} = \nu S_j^{(k-1)} - \lambda_{E,j}^{(k)} \quad (6)$$

with $\phi(F, R) \equiv F (1 - \exp(-R/R))$ (a “min-like” function that approaches F smoothly from below as $R \rightarrow \infty$ and *vice versa*) we previously defined $\lambda_{E,j}^{(k)}$ as $\min \left(\nu S_j^{(k-1)}, \rho c N_j^{(k,1)} \right)$

By Poisson thinning of the total propagule pool, the actual number of retained propagules $R_j^{(k)}$ and total exported propagules $E_j^{(k)}$ generated by patch j are:

$$R_j^{(k)} \sim \text{Poisson}(\lambda_{R,j}^{(k)}) \quad (7)$$

$$E_j^{(k)} \sim \text{Poisson}(\lambda_{E,j}^{(k)}) \quad (8)$$

Exported propagules are assumed to land uniformly at random among the n patches. The number of propagules exported from source patch j that arrive in target patch i is

$$Y_{j \rightarrow i}^{(k)} \sim \text{Poisson}\left(\frac{\lambda_{E,j}^{(k)}}{n}\right). \quad (9)$$

Summing over all sources gives the total number of imported propagules in patch i :

$$Y_{i,\text{import}}^{(k)} = \sum_{j=1}^n Y_{j \rightarrow i}^{(k)}. \quad (10)$$

Total propagules present in patch i The total number of propagules present in patch i at colonization is

$$G_i^{(k)} = R_i^{(k)} + Y_{i,\text{import}}^{(k)}. \quad (11)$$

4.3 Newly infected hosts

We now derive the per-host infection probability and the distribution of the number of newly infected hosts created during colonization.

Per-host infection probability. Fix a particular host in patch i . Given $G_i^{(k)}$ propagules, assume: (i) each propagule makes a $\text{Poisson}(\beta)$ number of effective bites; (ii) each bite chooses a host uniformly among the $N_i^{(k,2)}$ hosts in the patch; (iii) each bite infects with probability p , independently.

Then the number of *successful infection attempts* on the focal host is Poisson with mean

$$\frac{\beta p G_i^{(k)}}{N_i^{(k,2)}} = \frac{\alpha G_i^{(k)}}{N_i^{(k,2)}}.$$

Therefore, the probability that the focal host becomes infected is

$$\pi_i^{(k)} = 1 - \exp\left(-\frac{\alpha G_i^{(k)}}{N_i^{(k,2)}}\right) \approx \frac{\alpha G_i^{(k)}}{N_i^{(k,2)}}. \quad (12)$$

Number of newly infected hosts Given $\pi_i^{(k)}$, the number of hosts infected is

$$X_i^{(k)} \mid \left(G_i^{(k)}, N_i^{(k,2)}\right) \sim \text{Binomial}\left(N_i^{(k,2)}, \pi_i^{(k)}\right).$$

When $\pi_i^{(k)} \ll 1$, we can approximate this using the expectation of the total propagules present in the patch:

$$\begin{aligned}
X_i^{(k)} &\approx \text{Poisson}\left(N_i^{(k,2)} \pi_i^{(k)}\right) \\
&= \text{Poisson}\left(\alpha G_i^{(k)}\right) \\
&\approx \text{Poisson}\left(\alpha \left[\lambda_{R,i}^{(k)} + \frac{1}{n} \sum_{j=1}^n \lambda_{E,j}^{(k)} \right]\right)
\end{aligned} \tag{13}$$

Update infected count after colonization. We define

$$I_i^{(k,2)} = I_i^{(k,1)} + X_i^{(k)} = X_i^{(k)}. \tag{14}$$

5 Fizzle

Let $P_f = 1/\mathcal{R}_0$ be the probability that a single introduction fails to establish. If X introductions occur independently, then the probability that all of them fizzle is P_f^X . Therefore,

$$P(\text{establish}) = 1 - \mathbb{E}[P_f^X].$$

Since $X \sim \text{Poisson}(\lambda)$,

$$\mathbb{E}[P_f^X] = \sum_{x=0}^{\infty} P_f^x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda P_f)^x}{x!} = e^{-\lambda} e^{\lambda P_f} = e^{-\lambda(1-P_f)}.$$

Therefore

$$P(\text{establish}) = 1 - e^{-\lambda(1-P_f)}$$

where the expected number of successful primary infections λ is:

$$\lambda = \alpha \left[\lambda_{R,i}^{(k)} + \frac{1}{n} \sum_{j=1}^n \lambda_{E,j}^{(k)} \right]$$

Population size does not change:

$$N_i^{(k,3)} = N_i^{(k,2)}. \tag{15}$$

6 Major epidemic

In patches that experience a major epidemic, the population is reduced by the epidemic wave:

$$N_i^{(k,4)} = \left(1 - zD \mathbf{1}_i^{(k,3)}\right) N_i^{(k,3)}. \tag{16}$$

We assume infected rats die quickly and no infected rats remain at the end of a wave:

$$I_i^{(k,4)} = 0 \quad \text{for all } i. \tag{17}$$

7 Year-to-year update

$$N_i^{(k+1,0)} = N_i^{(k,4)}, \quad I_i^{(k+1,0)} = I_i^{(k,4)} = 0. \tag{18}$$